

Module/Pathway/Taxon Review: Bacterial Chemotaxis Signal Transduction to Flagellar Motor Output in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target pathway/bucket:** KEGG ppu02030 — Bacterial chemotaxis (module area: transport_motility_signaling) **Companion bucket:** KEGG ppu02040 — Flagellar assembly (motor switch + stator genes) **Candidate genes assessed:** 48 loci from local metadata

1. Executive Summary

The chemoreceptor-to-flagellar-motor signaling module is **fully satisfiable** in *Pseudomonas putida* KT2440. Every canonical step of the generic module — chemoreceptor input, CheA autophosphorylation/phosphotransfer, CheY/CheZ response-regulator control, FliG/FliM/FliN switch output, and MotAB-powered motor rotation — is encoded in the KT2440 genome, and the core signaling cascade has direct target-strain evidence. The KT2440 flagellar/chemotaxis gene cluster is experimentally defined as a single large region (≈ 59 genes in ≈ 11 operons), the core Che genes are organized in the *cheYZA* and *cheB-motAB* operons, and the whole apparatus is under a three-tier $\text{FleQ} \rightarrow \sigma^{54} \rightarrow \sigma^{28}$ (FliA) regulatory cascade with a confirmed nonmotile *fliA* phenotype. This is far stronger than homology-only satisfaction.

The chemoreceptor input step is the best-characterized part of the module in this organism. KT2440 is predicted to encode **27 chemoreceptors (MCPs)**, and the Krell laboratory generated the complete set of chemoreceptor deletion mutants. At least **seven receptors have directly determined ligand specificities in KT2440 or closely related *P. putida*:** McpS (TCA-cycle intermediates + butyrate), McpQ (citrate/metal complexes), McpP (C2/C3 carboxylic acids), McpU (polyamines), McpA (amino acids), McpG (GABA), and PcaY_PP (aromatic/hydroaromatic acids). These are high-confidence, promotion-worthy annotations.

Two boundary corrections are required for accurate curation. **First**, the motor-output step is incomplete as modeled: *P. putida* uses **two non-redundant stators, MotAB and MotCD**, and the candidate metadata captures only MotAB — the module should be marked `module_needs_revision`. **Second**, several candidate genes are **likely KEGG over-propagations**: the ABC-transporter periplasmic substrate-binding proteins (dppA-I..IV, rbsB, PP_2757/PP_2758) and Wsp/c-di-GMP chemosensory-like MCP loci do not represent flagellar chemotaxis-satisfying components in this organism and should be flagged `candidate_uncertain`. The CtpH/CtpL phosphate receptors rest on *P. aeruginosa* evidence (with CtpL working via a PstS binding-protein shuttle) and remain `candidate_uncertain` pending direct KT2440 data.

2. Target-Organism Pathway Definition

2.1 What the module includes

The module is the **flagellar chemotaxis signal-transduction pathway**: the information flow from transmembrane methyl-accepting chemotaxis proteins (MCPs), through the CheA/CheW signaling array and the CheY response regulator, to the FliM/FliN/FliG switch complex that biases the direction of rotation of the MotAB/MotCD-powered flagellar motor. Adaptation/reset components (CheR methyltransferase, CheB methylesterase, CheZ phosphatase, and CheV) are part of the pathway because they set the dynamic operating point of the receptor array and terminate the CheY-P signal.

In KT2440 the behavioral readout is a "**run-reverse-turn**" swimming pattern typical of polar-flagellated pseudomonads, rather than the enterobacterial "run-and-tumble" ([PMID: 23728820](#)). The motor spends roughly equal time in CW and CCW rotation, which is mechanistically relevant to how CheY-P bias is interpreted at the switch.

2.2 Neighboring pathways that must be kept separate

- **KEGG ppu02040 (Flagellar assembly)**. The structural flagellar genes (*fliM*, *fliN*, *fliG*, *motA*, *motB*, and the PP_4335/PP_4336 cluster-I motor genes) are annotated to ppu02040. The switch and stator components sit at the interface and are legitimately shared, but flagellar **biosynthesis** proper is a distinct map.
- **The Wsp chemosensory-like c-di-GMP pathway**. This is a *separate* signal-transduction system for surface sensing/biofilm, not flagellar chemotaxis. Its receptor WspA initiates a

phosphorylation cascade activating the diguanylate cyclase WspR, producing c-di-GMP rather than controlling motor bias (PMID: 35476526). KT2440 possesses a WspA homologue that controls biofilm (PMID: 26662997). Some MCP-like loci in the candidate list belong to this Wsp/F-type cluster and must not be counted toward flagellar chemotaxis.

- **c-di-GMP → motility modulation via FlgZ/YcgR-type control.** In *P. putida*, PP4397/FlgZ links c-di-GMP signaling (from PP2258) to flagellar motility (PMID: 30111852). This is a second-messenger modulation layer, not the core Che phosphorelay.

2.3 Alternate names / database definitions

KEGG "Bacterial chemotaxis" (map02030 / ppu02030) is the canonical reference. The pathway is also captured in GO under chemotaxis / signal transduction and taxis terms and in the module framework as a reusable "chemoreceptor-to-flagellar-motor signaling" module. Receptor gene names vary by database (systematic PP_ locus tags vs. functional names such as McpS, McpP, McpU, McpG, PcaY).

3. Expected Step Model

The generic module and its known step relationships map onto KT2440 as follows:

- [1] Chemical stimulus detection
MCP chemoreceptors (≈27 in KT2440) — ligand binding at LBD → promotes
- [2] Histidine-kinase phosphotransfer
CheA (PP_4338) autophosphorylation, via CheW (PP_4332/PP_4333) coupling → feeds into
- [3] Response-regulator control + reset
CheY (PP_4340) phosphorylation state ← CheZ (PP_4339) phosphatase reset
Adaptation: CheR (PP_4392/PP_3760), CheB (PP_4337/PP_3759), CheV (PP_2128) → causes
- [4] Motor-switch response
FliG (PP_4368) / FliM (PP_4358) / FliN (PP_4357) switch → causes
- [5] Proton-driven motor output
MotAB (PP_4905/PP_4904) + MotCD (SECOND STATOR – not in metadata)

Step	Expected player	KT2440 candidate	Status
1. Stimulus detection	MCP family	27 receptors; 7 characterized	covered (input richly documented)
2. CheA + coupling	CheA histidine kinase; CheW	cheA PP_4338; CheW PP_4332/PP_4333; CheV PP_2128	covered
3a. CheY response regulator	CheY	cheY PP_4340	covered
3b. CheZ reset	CheZ phosphatase	cheZ PP_4339	covered
3c. Adaptation	CheR / CheB	cheR2 PP_4392, cheR3 PP_3760, cheB1 PP_4337, PP_3759	covered (paralog ambiguity)
4. Switch	FliG/FliM/FliN	fliG PP_4368, fliM PP_4358, fliN PP_4357	covered (ppu02040)
5. Motor output	MotA/MotB	motA PP_4905, motB PP_4904 (+ PP_4335/PP_4336)	module_needs_revision (MotCD missing)

4. Candidate Genes and Evidence

4.1 Chemoreceptor input — high-confidence, target-strain evidence

KT2440 is repeatedly described as *"predicted to have 27 chemoreceptors, most of which uncharacterized"* (PMID: 26463109), and the complete set of chemoreceptor deletion mutants has been generated: *"Here we report the generation of the complete set of chemoreceptor mutants of Pseudomonas putida KT2440"* (PMID: 26662997). The candidate list contains ~30 MCP/aerotaxis loci, consistent with — but not identical to — this experimentally curated repertoire (some candidate MCP-like loci belong to Wsp/other chemosensory clusters).

Directly characterized receptors (promote to full review):

Receptor	Locus	Ligands	Evidence	PMID
McpS	PP_4658	6 TCA-cycle intermediates + butyrate; citrate–metal discrimination	ITC + chemotaxis, KT2440	20498372 , 21360620
McpQ	PP_5020	citrate/metal-ion complexes (McpS paralogue)	Direct, KT2440	26463109
McpP	PP_2861	acetate, pyruvate, propionate, L-lactate (KD 34–107 μ M)	ITC, KT2440	26048936
McpU	PP_1228	putrescine, cadaverine, spermidine (first polyamine receptor)	ITC + structure (dCACHE)	29758259
McpA	PP_2249	12 proteinogenic amino acids	Direct, KT2440	26662997
McpG	PP_1371	GABA (specific); root-colonization phenotype	ITC + chimera + Δ mcp, KT2440	25921834
PcaY_PP	PP_2643	aromatic/hydroaromatic acids (broad: C6 ring + carboxyl)	Structure + binding, KT2440	33021055

Key direct-evidence quotes: - McpS: "We report the identification of McpS as the specific chemoreceptor for 6 tricarboxylic acid (TCA) cycle intermediates and butyrate in *Pseudomonas putida*" (PMID: [20498372](#)). - McpP: "we identify PP2861 (termed McpP) of *Pseudomonas putida* KT2440 as a chemoreceptor with a novel ligand profile ... recognizes acetate, pyruvate, propionate, and l-lactate" (PMID: [26048936](#)). - McpU: "the McpU chemoreceptor from *Pseudomonas putida* was identified as the first chemoreceptor that bound specifically polyamines" (PMID: [29758259](#)). - McpG: "We report the identification of McpG as a specific GABA chemoreceptor in non-pathogenic *Pseudomonas putida* KT2440" (PMID: [25921834](#)), with Δ mcpG reducing tomato-root colonization and GABA detected in root exudates. - PcaY_PP: "characterized by an unusually broad signal range, and minimal requisites for signal binding are the presence of a C6-membered ring and that of a carboxyl group" (PMID: [33021055](#)).

Caveats for receptor curation: Many characterized *Pseudomonas* receptors are **promiscuous/multifunctional**. PcaY_PP has an "unusually broad signal range" (PMID: [33021055](#)); PcpI of *P. putida* responds to two different phytohormones (salicylate and IAA) without binding IAA directly (PMID: [35088505](#)). Receptor–ligand annotations should therefore

record a specificity *range*, not a single ligand, and should flag metabolism-independent vs. metabolism-coupled taxis (e.g., PcaY links chemotaxis, transport, and catabolism; [PMID: 25582673](#)).

4.2 Core signaling cascade — target-strain operon/regulatory evidence

The KT2440 flagella/chemotaxis cluster is experimentally defined. Rodríguez-Herva 2010 shows the cluster "*is comprised of four independent transcriptional units: flhAF, fleNfliA, cheYZA and cheBmotAB*" and that a nonpolar *fliA* (σ^{28}) mutant is nonmotile ([PMID: 23766109](#)). Leal-Morales 2022 shows the flagellar cluster is a single ≈ 59 -gene region in ≈ 11 operons, with the core chemotaxis machinery under a "*three-tier cascade in which fleQ is a Class I gene*" $\rightarrow \sigma^{54} \rightarrow \sigma^{28}$ (FliA) ([PMID: 34859548](#)).

Component	Locus	Role	Evidence tier
CheA (EC 2.7.13.3)	PP_4338	Histidine kinase, autophosphorylation	Operon-defined (<i>cheYZA</i>), KT2440
CheY	PP_4340	Response regulator; sets motor bias	Operon-defined, KT2440
CheZ (EC 3.1.3.-)	PP_4339	Phosphatase reset of CheY-P	Operon-defined, KT2440
CheB1 (EC 3.1.1.61/3.5.1.44)	PP_4337	Methylesterase (adaptation); <i>cheB-motAB</i> operon	Operon-defined, KT2440
CheR2 (EC 2.1.1.80)	PP_4392	Methyltransferase (adaptation)	Homology + operon context
CheW	PP_4332 / PP_4333	CheA-MCP coupling	Homology, cluster context
CheV-like	PP_2128	CheW-CheY hybrid coupling/adaptation	Homology

A *P. putida* chemoreceptor forms **active signaling complexes when reconstituted with *E. coli* CheA/CheW/CheY**, with methylation-dependent adaptation ([PMID: 22745269](#)), functionally confirming the receptor $\rightarrow$ kinase $\rightarrow$ CheY logic is conserved.

4.3 Switch and motor — covered, with a required revision

FliG (PP_4368), FliM (PP_4358), FliN (PP_4357), MotA (PP_4905), and MotB (PP_4904) cover the switch and one stator. However, **motor torque in *P. putida* is generated by two stators**: "*the motor torque for flagellar rotation is generated by the two stators MotAB and MotCD*" (PMID: 36409076). MotAB is essential for swimming in liquid; MotCD is required for spreading in semisolid agar — the two are **non-redundant**. The candidate metadata lists only MotAB (plus PP_4335/PP_4336 cluster-I motor genes) and does not flag the MotCD set. This is the single clearest module-boundary correction.

4.4 Aerotaxis / energy taxis receptors

PP_2111, PP_2257, PP_4521 are annotated as aerotaxis receptors. Energy/aerotaxis is a legitimate sub-branch of chemotaxis (Aer-type receptors feed the same CheA/CheY cascade). Phylogenetic and functional work on Aer receptors in pseudomonads supports their role in energy taxis (PMID: 31639075; PMID: 35264479). Direct KT2440 functional data are thinner than for the carboxylate/amino-acid receptors; treat as `covered` at the family level but `candidate_uncertain` for individual locus assignment.

5. Gaps, Ambiguities, and Likely Over-Annotations

5.1 Substrate-binding proteins — likely KEGG over-propagation

The candidate list includes **dppA-I..IV (PP_0882/0884/0885/5283), rbsB (PP_2454), and PP_2757/PP_2758** (ribose/sugar-binding), all assigned to kegg:ppu02030. These are ABC-transporter periplasmic **substrate-binding proteins (SBPs)**. KEGG map02030 includes SBPs because in *E. coli* certain SBPs (DppA, RbsB, MglB) hand ligands to the Tap/Trg MCPs; this ortholog assignment is auto-propagated to ppu02030. **No direct KT2440 evidence links these SBPs to chemoreceptor signaling**. The characterized KT2440 receptors bind ligands **directly via their own LBDs** (McpS, McpP, McpU, McpG, PcaY) rather than through SBP intermediaries. → Flag `candidate_uncertain` / over-propagation.

Calibration caveat. SBP-mediated chemotaxis is not impossible in *Pseudomonas*. In *P. aeruginosa*, the periplasmic Pi-binding protein **PstS** binds phosphate and then stimulates the CtpL receptor — a genuine SBP shuttle: "*We identify the periplasmic ligand binding protein PstS as the protein that binds in its Pi loaded state to CtpL, resulting in receptor stimulation. PstS*

forms part of the Pi transporter and has thus a double function in Pi transport and chemotaxis" (PMID: 27353565). So the mechanism exists in the genus; the flag is "unsupported for these specific loci in KT2440," not "impossible."

5.2 CtpH/CtpL phosphate chemotaxis — species-transfer uncertainty

ctpH (PP_2120) and ctpL (PP_0562) are named by homology. The phosphate-taxis evidence originates in *P. aeruginosa*: "Two chemotactic transducers for inorganic phosphate (Pi), designated CtpH and CtpL, have been identified in *Pseudomonas aeruginosa*" (PMID: 10852870), with CtpH binding Pi directly and CtpL requiring the PstS shuttle (PMID: 27353565). No direct KT2440 phosphate-taxis data were found. Species transfer is **plausible but unverified** → candidate_uncertain.

5.3 Wsp / c-di-GMP chemosensory-like loci — wrong pathway

Some MCP-like loci in the candidate set belong to the **Wsp (c-di-GMP/biofilm)** system, which is distinct from flagellar chemotaxis: "The methyl-accepting chemotaxis protein WspA recognizes an unknown surface-associated signal and initiates a phosphorylation cascade that activates the diguanylate cyclase WspR" (PMID: 35476526; see also PMID: 26662997). Related Wsp-like systems in other bacteria confirm this architecture (e.g., *Halomonas* HtChe2/Htc10 controlling a diguanylate cyclase; PMID: 39529381). These loci should not be counted as flagellar chemotaxis-satisfying → candidate_uncertain / reassign.

5.4 CheR/CheB paralog ambiguity

KT2440 encodes multiple adaptation-enzyme paralogs: CheR2 (PP_4392) and CheR3 (PP_3760); CheB1 (PP_4337) and a second methylesterase PP_3759. The broad EC mappings (EC 3.1.1.61, EC 2.1.1.-) and multiple paralogs mean locus-level assignment to "the" flagellar Che pathway vs. auxiliary chemosensory clusters needs case-by-case curation. The *cheB-motAB* operon anchors CheB1/PP_4337 to the flagellar cascade with high confidence; the others are candidate_uncertain.

5.5 CheW duplication

PP_4332 (CheW) and PP_4333 (CheW domain protein) are adjacent paralogs. Both plausibly participate in array coupling; assignment of distinct roles (or redundancy) is unresolved.

6. Module and GO-Curation Recommendations

Module step	Recommended status	Rationale
Chemoreceptor input	covered	27 receptors; 7 with direct ligand data (McpS, McpQ, McpP, McpU, McpA, McpG, PcaY_PP)
CheA autophosphorylation/ phosphotransfer	covered	cheA PP_4338, operon-defined (<i>cheYZA</i>); reconstituted signaling shown
CheY response regulator	covered	cheY PP_4340, operon-defined
CheZ reset	covered	cheZ PP_4339, operon-defined
Adaptation (CheR/CheB/ CheV)	covered (with paralog caveats)	CheB1/PP_4337 operon-anchored; paralogs candidate_uncertain
FliG/FliM/FliN switch	covered	ppu02040 loci present
MotAB stator	covered	motA/motB present
Second stator (MotCD)	module_needs_revision	<i>P. putida</i> has two non-redundant stators; MotCD absent from metadata
SBPs (dppA, rbsB, PP_2757/2758)	candidate_uncertain / over-annotation	KEGG auto-propagation; no KT2440 chemotaxis link
CtpH/CtpL phosphate taxis	candidate_uncertain	<i>P. aeruginosa</i> evidence; species transfer unverified
Wsp/c-di-GMP MCP-like loci	candidate_uncertain / reassign	Belongs to biofilm c-di-GMP system, not flagellar output

Module boundary revision needed: The generic module's single "MotAB stator-driven flagellar rotation" step should be generalized to a **dual-stator (MotAB + MotCD)** motor-output step for *P. putida* and related pseudomonads, capturing the non-redundant liquid-vs-surface division of labor.

Possible new module/GO requests: - A separate **Wsp / c-di-GMP surface-sensing chemosensory module** to receive the MCP-like loci that are currently mis-bucketed into ppu02030. - A **c-di-GMP → FlgZ/YcgR motility-modulation** annotation to capture PP4397/FlgZ ([PMID: 30111852](#)) as a distinct regulatory layer. - GO annotation extensions recording receptor **ligand-range** rather than single-ligand specificity for promiscuous receptors (PcaY_PP, PcpI).

7. Genes to Promote to Full fetch-gene Review

High priority (direct KT2440 functional data — confirm annotations and record ligand ranges): - McpS (PP_4658), McpQ (PP_5020), McpP (PP_2861), McpU (PP_1228), McpA (PP_2249), McpG (PP_1371), PcaY_PP (PP_2643) — characterized receptors. - cheA (PP_4338), cheY (PP_4340), cheZ (PP_4339), cheB1 (PP_4337) — operon-anchored core cascade. - motA (PP_4905), motB (PP_4904) — resolve dual-stator status; locate/annotate the MotCD stator pair.

Medium priority (ambiguity/over-annotation to resolve): - ctpH (PP_2120) / ctpL (PP_0562) — verify or downgrade phosphate-taxis assignment for KT2440. - cheR2 (PP_4392) / cheR3 (PP_3760) / PP_3759 — resolve which paralogs serve the flagellar cascade. - PP_4332 / PP_4333 — resolve CheW paralog roles. - dppA-I..IV, rbsB (PP_2454), PP_2757/PP_2758 — confirm reassignment out of chemotaxis (transport only). - Aerotaxis loci PP_2111, PP_2257, PP_4521 — confirm energy-taxis role and cluster membership. - Uncharacterized MCP loci (PP_0317, PP_0320/mcpH, PP_0584, PP_0779, PP_0802, PP_1819, PP_1940, PP_2823, PP_3414, PP_3557, PP_5021) — triage Che vs. Wsp cluster membership.

8. Limitations and Knowledge Gaps

1. **MotCD locus not identified in metadata.** The dual-stator conclusion is from the literature ([PMID: 36409076](#)); the specific PP_ loci for MotCD were not resolved in this review and should be located during curation.
2. **~20 of 27 receptors remain functionally uncharacterized**, so input-step coverage is documented at the family level, with specific ligand assignments for only 7.
3. **Cluster membership of many MCP-like loci is not individually resolved** (Che flagellar vs. Wsp c-di-GMP vs. other chemosensory arrays). This is the largest single source of over-annotation risk.

4. **CtpH/CtpL and several other assignments rest on homology transfer from *P. aeruginosa* or *P. putida* F1**, not direct KT2440 experiments.
 5. Receptor promiscuity means single-ligand annotations will understate true specificity ranges.
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9. Proposed Follow-up Experiments / Actions

- **Curation:** Locate the MotCD stator loci in KT2440 and add a dual-stator motor-output step to the module. Reassign SBP and Wsp loci out of ppu02030.
 - **Bioinformatic triage:** Classify all 27 MCP-like loci by chemosensory-cluster membership (Che vs. Wsp vs. Aer) using operon context and array-type signatures, resolving the candidate list against the experimentally curated 27-receptor set.
 - **Targeted experiments (if wet-lab follow-up is possible):** ITC/chemotaxis screens for the uncharacterized MCPs; direct phosphate-taxis assay in KT2440 to test CtpH/CtpL; motility phenotyping of single/double stator mutants in liquid vs. semisolid to confirm MotAB/MotCD division of labor.
 - **Expert questions:** Which CheR/CheB paralogs and CheW copies are dedicated to the flagellar array vs. auxiliary systems? Does KT2440 use SBP-shuttle chemotaxis for any ligand (as *P. aeruginosa* does via PstS)?
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10. Key References

PMID	Contribution
26463109	27-receptor count; McpQ citrate specificity
26662997	Complete KT2440 receptor mutant set; McpA; biofilm
20498372	McpS = TCA-cycle intermediate receptor
21360620	McpS citrate-metal discrimination
26048936	McpP = C2/C3 carboxylic-acid receptor
29758259	McpU polyamine receptor structure
33021055	PcaY_PP broad aromatic specificity
25921834	McpG = GABA receptor; root colonization
25582673	PcaY links chemotaxis/transport/catabolism
35088505	PcpI dual-phytohormone (receptor promiscuity)
22745269	<i>P. putida</i> receptor signals in <i>E. coli</i> Che complex
23766109	KT2440 <i>cheYZA/cheB-motAB</i> operons; <i>fliA</i> nonmotile
34859548	FleQ $\rightarrow\sigma^{54}\rightarrow$ FliA regulatory cascade
36409076	Two stators MotAB + MotCD
23728820	Run-reverse-turn motility paradigm
35476526	Wsp c-di-GMP pathway distinct from chemotaxis
30111852	FlgZ c-di-GMP motility modulation
27353565	PstS shuttle for CtpL (SBP calibration)
10852870	CtpH/CtpL phosphate taxis (<i>P. aeruginosa</i>)
39529381	Wsp-like DGC-controlling receptor (comparative)
24148021	Metabolic cost of flagellar motility in KT2440
31639075	Aer energy-taxis receptor phylogeny

Review scope note: Conclusions on the core cascade, operon organization, regulation, and the seven characterized receptors are supported by **direct *P. putida* KT2440 (or closely related *P. putida*) experiments. The dual-stator architecture is direct *P. putida** evidence. CtpH/CtpL phosphate taxis and the SBP-shuttle mechanism derive from *P. aeruginosa**** and require verification before transfer to KT2440. SBP and Wsp over-annotation flags are inferences from pathway-database propagation logic combined with the absence of KT2440-specific chemotaxis evidence.**

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